

19. The most popular fertilizer for foliar application is –

- (a) Sodium Nitrate (b) Urea
(c) D.A.P. (d) Potassium Chloride

U.P. Lower Sub. (Pre) 2015
U.P.P.C.S. (Mains) 2015

Ans. (b)

The foliar fertilization means applying nutrients to plant leaves or needles. Urea is commonly used for foliar fertilization because of its high solubility and its rapid and effective absorption by leaves.

20. 'Humus' is an example of –

- (a) Soil structure (b) Crystalloids
(c) Organic colloids (d) None of the above

U.P.R.O./A.R.O. (Pre) 2014

Ans. (c)

Humus usually refers to the natural decay of material such as leaves in the soil's top layer. It is a dark organic material that forms in the soil when plant and animal matter decays.

21. The most suitable crop for green manuring is –

- (a) Potato (b) Sugarcane
(c) Sunhemp (d) Gram

U.P.P.C.S. (Mains) 2015

Ans. (c)

The green undecomposed material used as manure is called green manure. It is obtained in two ways, by growing green manure crops or by collecting green leaf (along with twigs) from plants grown in wastelands, field bunds and forests. Green manuring is growing in the field plants usually belonging to leguminous family and incorporating into the soil after sufficient growth. The plants that are grown for green manure known as green manure crops. The most important green manure crops are sunhemp, dhaincha, pillipesara, clusterbean and sesbania rostrata.

Miscellaneous

Notes

- **Allethrin** is a synthetic duplicate of a component of pyrethrum. Pyrethrum is a natural insecticide extracted from Chrysanthemum flower. It is a **mosquito repellent**.
- A **fire extinguisher** is an active fire protection device used to extinguish to control fires, often in emergency situations. In this apparatus generally carbon dioxide is produced by the reaction of sodium bicarbonate and dil. sulphuric acid.

$$2\text{NaHCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O} + 2\text{CO}_2$$
- **Rodenticides** are pesticides that kill rodents. It is also called rat poison. Rodenticides are made from the

chemicals such as potassium cyanide or zinc phosphide. These are very poisonous chemicals.

- **Aluminium phosphide** is a cheap, easily available agricultural pesticide which causes lethal poisoning by liberation of phosphine gas which causes cell hypoxia due to inhibition of oxidative phosphorylation, leading to circulatory failure. Aluminium phosphide is also used as **fumigant** for preservation of cereals.
- **Mustard Gas** is the prototypical substance of the sulphur-based family of cytotoxic and vesicant chemical warfare agents known as the sulfur mustards. Mustard gas can form large blisters on exposed skin and in the lungs. Mustard gas is a fatal chemical which affects extremely to the cells of skin, eye, lungs and DNA. Mustard gas was used in first World War as chemical weapon. Its chemical name is dichloro diethyl sulphide.
- **Uranium dating** is one of the ways of determining the age of ancient objects such as earth and rocks.
- **Carbon dating** is a method for the determining the age of an object containing organic material.
- **Crookes glass** is made by mixing of **cerium** and some other rare earth metals. Crookes glass has capacity to absorb the ultraviolet rays. It is used to make sunglasses.
- **Soda-lime glass** is produced on a large scale and used for bottles, drinking glasses, and windows. Its light transmission properties, as well as low melting temperature, make it suitable for use as window glass. Its smooth and non-reactive surface makes it excellent as containers for food and drinks.
- **Jena glass** is a shock-and heat-resistant glass used in scientific and technological applications, especially in chemistry.
- **Pyrex glass** is a low-thermal-expansion borosilicate glass used for laboratory glassware and kitchenware.
- **Zinc Oxide (ZnO)** is also known as **flowers of zinc**, a white powder used as a pigment in manufacturing of cosmetics, inks and ointments.
- **Cloud seeding** is the process of spreading either dry ice or more commonly, **silver iodide aerosols**, into the upper part of the clouds to try to stimulate the precipitation process and form rains. These chemicals are sprayed over clouds by planes. The above chemicals are also shoot from the ground by generators or anti-aircraft guns.
- **Phosphorus oxide** is responsible for the inflammability of match. The chemical present on the tip of match stick is **potassium chlorate**, **sulphur**, **starch** and **glue**. Match box striking surface contains **red phosphorus**, **powdered glass** and **glue**.
- **Nickel and Cadmium** pairs of material serves as electrodes in chargeable batteries commonly used in devices such as torch light and electric shaver.

Question Bank

1. In the year 2021, the Nobel Prize in Chemistry was awarded to :

- (a) Syukuro Manabe, Klaus Hasselmann & Giorgio Parisi
- (b) David Julius & Ardem Patapoutian
- (c) Benjamin List & David MacMillan
- (d) Abdulrazak Gurnah

Uttarakhand P.C.S. (Pre) 2021

Ans. (c)

The Nobel Prize in Chemistry, 2021 was awarded jointly to Benjamin List and David W.C. MacMillan for the development of asymmetric organocatalysis. The Nobel Prize in Chemistry, 2022 was awarded jointly to Carolyn R. Bertozzi, Morten Meldal and K. Barry Sharpless for the development of click chemistry and bioorthogonal chemistry. Mouni G. Bawendi, Louis E. Brus and Aleksey Yekimov were awarded the Nobel Prize in Chemistry, 2023 for the discovery and synthesis of quantum dots. The Nobel Prize in Chemistry, 2024 was awarded with one half to David Baker for computational protein design and the other half jointly to Demis Hassabis and John Jumper for protein structure prediction.

2. The Nobel Prize in Chemistry, 2019 was awarded for the development of :

- (a) LED
- (b) LCD
- (c) Lithium-ion batteries
- (d) MRI
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

The Nobel Prize in Chemistry, 2019 was awarded to John B. Goodenough, M. Stanley Whittingham and Akira Yoshino for the development of lithium-ion battery. The Nobel Prize in chemistry, 2020 was awarded to Emmanuelle Charpentier and Jennifer A. Doudna for the development of a method for genome editing (CRISPR/Cas 9 genetic scissors).

3. Which of the following types of glass can cut off ultra-violet rays?

- (a) Soda glass
- (b) Pyrex glass
- (c) Jena glass
- (d) Crookes glass

U.P. P.C.S. (Pre) 2021

I.A.S. (Pre) 2007

Ans. (d)

- ♦ **Crookes glass** is one of the several types of glasses designed to diminish the transmission of ultraviolet rays. It is used to make sunglasses. It is produced by mixing mainly cerium oxide with silica.
- ♦ **Pyrex glass** is a clear, low-thermal-expansion borosilicate glass used for laboratory glassware and kitchenware.

- ♦ **Jena glass** is a shock- and heat-resistant glass used in scientific and technological applications, especially in chemistry.

- ♦ **Soda-lime glass** is produced on a large scale and used for bottles, drinking glasses, and windows.

4. Consider the following :

- | | |
|--------------------|----------------|
| 1. Aerosols | 2. Foam agents |
| 3. Fire retardants | 4. Lubricants |

In the making of how many of the above are hydrofluorocarbons used?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

I.A.S. (Pre) 2023

Ans. (c)

Hydrofluorocarbons (HFCs) are a group of synthetic gases primarily used for cooling and refrigeration. HFCs are entirely man-made. They are primarily produced for use in refrigeration, air-conditioning, insulating foams and aerosol propellants, with minor uses as solvents and for fire protection. HFCs are not used to make lubricants. Although, various types of aerosols (e.g. lubricant sprays and air dusters) require non-flammable propellants, and currently, HFCs are used in them. Hence, option (c) is the correct answer.

5. Ozone layer of upper atmosphere is being destroyed by which of the following?

- (a) Sulphur Dioxide
- (b) Photo-chemical oxidants
- (c) Chlorofluorocarbons
- (d) Smog

U.P. R.O./A.R.O. (Mains) 2017

Ans. (c)

When chlorine and bromine atoms make contact with ozone in the stratosphere, they destroy ozone molecules. One chlorine atom of chlorofluorocarbon can destroy over 100,000 ozone molecules before it is removed from the stratosphere. Ozone can be destroyed more quickly than it is naturally created.

6. Consider the following assertion and reason and select the correct answer from the code given below.

Assertion (A) : Ozone kills microorganism in water.

Reason (R) : Ozone is not radioactive.

Codes :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

U.P.P.C.S. (Mains) 2013

Ans. (b)

Ozone is made up of three oxygen atoms (O_3) having a 'free radical' of oxygen. It will readily give up one atom of oxygen providing a powerful oxidizing agent which is toxic to most waterborne organisms such as bacteria, mold and yeast spores, viruses or harmful protozoans that form cysts. Ozone is not radioactive. Thus, Both assertion and reason are true but reason is not the correct explanation of assertion.

7. Which of the following is used in fire alarm?

- (a) Photocell (b) Water
(c) Solar cell (d) None of the above

U.P. P.C.S. (Pre) 2020

Ans. (a)

The photocell is used in fire alarm. A photocell is a technological application of the photoelectric effect. A photocell is a resistor that changes resistance depending on the amount of light incident on it. It operates on semiconductor photoconductivity - the energy of light photons hitting the semiconductor frees electron to flow, decreasing the resistance. Photocell fire alarm device projects a continuous, focused beam of light onto a mirror. If nothing interferes with the beam en route and back, the sensor perceives no obstruction. But when a small amount of smoke enters the chamber, the light is refracted slightly, causing an alarm to sound.

8. Which gas is used in fire extinguisher?

- (a) Carbon dioxide (b) Hydrogen
(c) Oxygen (d) Sulphur dioxide

M.P.P.C.S. (Pre) 1991

Ans. (a)

Carbon dioxide gas is used in a fire extinguisher to control small fires, often in emergency situations.

9. In the ordinary fire extinguisher, Carbon dioxide is generated by the reaction of :

- (a) Limestone and dilute Sulphuric acid
(b) Marble powder and dilute Hydrochloric acid
(c) Sodium bicarbonate and dilute Sulphuric acid
(d) Magnesite and dilute Hydrochloric acid

Uttarakhand Lower Sub. (Pre) 2010

Ans. (c)

In the ordinary fire extinguisher, carbon dioxide is generated by the reaction of sodium bicarbonate and dilute sulphuric acid.
 $2NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O + 2CO_2$

10. Which of the following are the constituents of a soda acid fire extinguisher ?

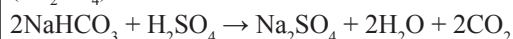
- (a) Conc. sulphuric acid and aluminium sulphate

- (b) Dil. sulphuric acid and sodium carbonate
(c) Dil. sulphuric acid and sodium bicarbonate solution
(d) Conc. sulphuric acid and sodium carbonate solution

70th B.P.S.C. (Pre) 2024

Ans. (c)

A soda-acid fire extinguisher contains a solution of sodium bicarbonate (baking soda) and sulphuric acid, which, when mixed, produce carbon dioxide gas that helps extinguish fires by displacing oxygen. Sodium bicarbonate ($NaHCO_3$) is the 'soda' part of the extinguisher, while sulphuric acid (H_2SO_4) is the 'acid' part, and it's a diluted solution of sulphuric acid. When the sulphuric acid comes into contact with the sodium bicarbonate solution, a chemical reaction occurs, producing carbon dioxide (CO_2), water (H_2O), and sodium sulphate (Na_2SO_4).



The carbon dioxide gas produced by the reaction displaces oxygen, which is needed for combustion, thus smothering the fire.

11. Which one of the following compounds is used to make fireproof clothing?

- (a) Sodium Sulphate (b) Magnesium Sulphate
(c) Ferrous Sulphate (d) Aluminium Sulphate

U.P.P.C.S. (Mains) 2015

Ans. (d)

Aluminium sulphate is used to prepare fireproof clothes.

12. An active chemical to remove mosquitoes is—

- (a) Allethrin
(b) Atropin
(c) 2-ISO propoxiphynyl
(d) Benzene Hexachlorophene

41st B.P.S.C. (Pre) 1996

Ans. (a)

Allethrin is a pyrethroid synthesized insecticide/pesticide. It is an active chemical to remove mosquitoes.

13. Prallethrin is a common :

- (a) Analgesic (b) Muscle relaxant
(c) Mosquito repellent (d) Hair dye

U.P. P.C.S. (Mains) 2017

Ans. (c)

Prallethrin is a pyrethroid insecticide. Prallethrin 1.6% w/w liquid vaporizer is a repellent insecticide which is generally used for the control of mosquitoes in the household. It is marketed as a mosquito repellent by Godrej as 'Good Knight Silver Power' and SC Johnson as 'All out' in our country.

14. Which of the following is used as a mosquito repellent?

- (a) Pyrethrum
- (b) Rotenone
- (c) Ephedrine
- (d) None of the above

U.P.P.C.S. (Mains) 2009

Ans. (c)

Pyrethrum is a natural botanical insecticide. It has potent insecticidal activity by targeting the nervous systems of insects which affect the nervous system of mosquitoes and diverge them.

15. Pyrethrin used in mosquito coil is obtained from –

- (a) A seed plant
- (b) An insect
- (c) A bacterium
- (d) A fungus

U.P.P.C.S. (Pre) 2005

Ans. (a)

Pyrethrins are a class of organic compounds normally derived from Chrysanthemum cinerariifolium, a seed plant that has potent insecticidal activity by targeting the nervous system of insects.

16. Which one of the following is used in preparing anatural mosquito repellent?

- (a) Congress grass
- (b) Elephant grass
- (c) Lemongrass
- (d) Nut grass

I.A.S. (Pre) 2021

Ans. (c)

Lemongrass, also known as citronella grass or fever grass, is a genus of Asian, African, Australian, and tropical island plants in the grass family. It is used for the production of citronella oil, which is used in soaps, as an insect repellent (especially for mosquitoes and houseflies), in insect sprays and candles, and in aromatherapy. It is known to be an excellent mosquito repellent. Around 95% of mosquitos from a specific range of species are either killed or completely repelled due to the intense odour of lemongrass oil.

17. Which chemical substance is used for making rat poison?

- (a) Ethyl Alcohol
- (b) Methyl Isocyanate
- (c) Potassium Cyanide
- (d) Ethyl Isocyanide

47th B.P.S.C. (Pre) 2005

Ans. (c)

Potassium cyanide (KCN) and zinc phosphide which are highly toxic chemicals, used as a rodenticide - a poison to kill the mouse.

18. Which one of the following is used as rat poison ?

- (a) Zinc Sulphide
- (b) Lead Sulphide
- (c) Calcium Phosphate
- (d) Zinc Phosphide

U.P.P.C.S. (Mains) 2012

Ans. (d)

See the explanation of above question.

19. Rodenticide is the chemical which is used to control :

- (a) Bats
- (b) Monkeys
- (c) Mouse
- (d) Rabbits

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

See the explanation of above question.

20. Aluminium Phosphide is used as –

- (a) Insecticide
- (b) Fungicide
- (c) Rodenticide
- (d) Fumigant

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (*)

Aluminium Phosphide is highly toxic inorganic compound. It is a cheap, easily available pesticide which causes lethal poisoning by liberation of phosphine gas. It is also used as fumigant for preservation of cereals and as rodenticide.

21. Which of the following is an example of herbicide?

- (a) Sodium chlorate
- (b) Potassium permanganate
- (c) Bleaching powder
- (d) None of the above

U.P.P.C.S. (Pre) 2019

Ans. (a)

Herbicides, also commonly known as weedkillers, are substances used to control unwanted plants. Sodium chlorate (NaClO_3) is an oxidizing agent and it is used as herbicide.

22. In India, the use of carbofuran, methyl parathion, phorate and triazophos is viewed with apprehension. These chemical are used as :

- (a) pesticides in agriculture
- (b) preservatives in processed foods
- (c) fruit-ripening agents
- (d) moisturising agents in cosmetics

I.A.S. (Pre) 2019

Ans. (a)

Carbofuran, methyl parathion, phorate and triazophos are used as pesticides in agriculture. On the recommendation of Anupam Verma Committee, Union Ministry of Agriculture and Farmers Welfare had notified Pesticides (Prohibition) Order, 2018 under which use of 18 pesticides was banned (12 from 8 August, 2018 and rest 6 from 31 December, 2020). This list include methyl parathion, phorate and triazophos.

23. The Weedicide commonly used for paddy is :

- (a) DDT
- (b) Dalapon
- (c) 2, 4-D
- (d) Ammonium Sulfamate

U.P. Lower Sub. (Mains) 2015

Ans. (c)

The weedicide commonly used for paddy is 2,4-D (2,4-Dichlorophenoxyacetic acid). Its chemical formula is $C_8H_6Cl_2O_3$. It is selective translocated herbicide and most widely used to control broadleaf weeds.

24. 2, 4-D is :

- (a) An insecticide (b) An explosive
(c) A fungicide (b) A herbicide

U.P.P.C.S. (Pre) 1999

Ans. (d)

2, 4-Dichlorophenoxyacetic acid (usually referred to by its abbreviation, 2,4-D) is a common systemic herbicide used in the control of broadleaf weeds. It is one of the most widely used herbicides in the world. 2,4-D is a synthetic auxin (plant hormone) and it is often used in laboratories for plant research and as a supplement in plant cell culture media such as MS medium.

25. Dichloro-diphenyl-trichloroethane (DDT) is a :

- (a) Non-biodegradable pollutant
(b) Biochemical pollutant
(c) Non-pollutant
(d) Biodegradable pollutant

U.P. P.C.S. (Pre) 2022

Ans. (a)

Dichlorodiphenyltrichloroethane (DDT) is an insecticide used in agriculture. It is an organic halogen that acts by disrupting the function of the nervous system, and kills insects rapidly upon contact. DDT is harm to all biota of the planet as it is non-biodegradable, and tends to become more concentrated as it moves up the food chain. It is a fat-soluble chemical, and accumulates in the fat deposits of the body. The United States banned the use of DDT in 1972. Some countries outside the United States still use DDT to control of mosquitoes that spread malaria. DDT and its related chemicals persist for a long time in the environment and in animal tissues.

26. Which one of the following substances is a herbicide?

- (a) Chlorpyrifos (b) Carbendazim
(c) Quinolphos (b) Butachlor

U.P. R.O./A.R.O. (Pre) 2017

Ans. (d)

Butachlor is a herbicide of the acetanilide class. It is extensively used in the form of granules in rice farming.

27. Isotope ^{60}Co is associated with :

- (a) For cancer diagnosis
(b) Determining the age of the earth
(c) Determining the age of fossil

(d) For trailer technology

R.A.S./R.T.S. (Pre) 1993

Ans. (c)

Radioisotope Carbon-14 (^{14}C) is used to determine the age of wood and bones (the time which elapsed after the death of plant or animal fossil) which is called Radiocarbon Dating. This technique is based on the decay of the Carbon-14 isotope in dead organism. The half-life of Carbon-14 is $5,730 \pm 40$ years.

28. Which of the following archaeological samples is adopted to determine the age of firewood bone and shell?

- (a) Uranium-238 (b) Argon-Isotopes
(c) Carbon-14 (d) Strontium-90

I.A.S. (Pre) 1993

Ans. (c)

See the explanation of above question.

29. Carbon dating is used to determine the age of :

- (a) Trees (b) Earth
(c) Fossils (d) Rocks

M.P.P.C.S. (Pre) 1993

Ans. (c)

See the explanation of above question.

30. 'Radioactive dating' is a technique which can be used to measure :

- (a) The age of rocks (b) Composition of rocks
(c) Colour of rocks (d) Weight of rocks

U.P.P.C.S. (Pre) 2008

Ans. (a)

Radioactive dating is a technique used to date material such as rocks or carbon. It is a principal source of information about the absolute age of rocks and other geological features.

31. Which of the following method is used for measuring the age of the earth?

- (a) Carbon dating process (b) Biotechnology Law
(c) Biological clock method (d) Uranium process

U.P.P.C.S. (Pre) 1992

Ans. (d)

Uranium-lead dating is one of the oldest and most refined methods to determine the age of the earth, decaying rocks and bones.

Clair Cameron Patterson, an American geochemist who pioneered studies of Uranium-lead radiometric dating methods is famous for having used to obtain one of the earliest accurate estimates of the age of the Earth.

32. One carbon credit is equivalent to :

- (a) 10 kg of CO₂ (b) 100 kg of CO₂
(c) 1000 kg of CO₂ (d) 10000 kg of CO₂

Uttarakhand P.C.S. (Pre) 2012

Ans. (c)

The idea for carbon credits came from the Kyoto Protocol of 1997. A carbon credit is a tradable permit or certificate that provides the holder of the credit the right to emit one metric ton (1,000 kg or 2,204 pounds) of carbon dioxide or an equivalent mass of another greenhouse gas. Carbon credits and carbon markets are a component of national and international attempts to mitigate the growth in concentration of greenhouse gases.

33. With which one of the following is BRIT (Government of India) engaged?

- (a) Railway Wagons (b) Information Technology
(c) Isotope Technology (d) Road Transport

I.A.S. (Pre) 2008

Ans. (c)

BRIT – Board of Radiation and Isotope Technology is an industrial unit of Department of Atomic Energy (DAE), Government of India, which provides the services based on radiation and isotopes for application in healthcare, agriculture, research and industry etc. Its headquarter is located at Vashi, Navi Mumbai.

34. Which one among the following is called philosopher's wool?

- (a) Zinc Bromide (b) Zinc Nitrate
(c) Zinc Oxide (d) Zinc Chloride

U.P.P.C.S. (Pre) 2007

Ans. (c)

The compound Zinc Oxide (ZnO) is called philosopher's wool. Alchemists, as part of their rituals, would burn zinc in air and collect the residue, which formed into white woolly tufts. They called it lana philosophica in Latin, meaning philosopher's wool. Zinc oxide has many uses such as white pigment in paints, component in zinc ointment for treating skin diseases, material in sun screens and sun lotions, in rubber manufacture and in photocopying products.

35. A pungent smell often present near the urinals is due to :

- (a) Sulphur dioxide (b) Chlorine
(c) Ammonia (d) Urea

I.A.S. (Pre) 2007

Ans. (c)

Urine odour is caused by the presence of ammonia. Urine is an aqueous solution of about 91-96% water. Other constituents include approx. are : Urea 2.6%, Sodium 0.4%, Ammonia 0.05%, Phosphate 0.6%, Creatinine and other dissolved ions.

36. A team of scientists at Brookhaven National Laboratory including those from India created the heaviest antimatter (antihelium nucleus). What is/are the implication/implications of the creation of anti-matter?

1. It will make mineral prospecting and oil exploration easier and cheaper.
2. It will help probe the possibility of the existence of stars and galaxies made of antimatter.
3. It will help to understand the evolution of the universe.

Select the correct answer using the codes given below.

- (a) Only 1 (b) 2 and 3
(c) Only 3 (d) 1, 2 and 3

I.A.S. (Pre) 2012

Ans. (b)

The scientists have discovered the heaviest antimatter ever observed in 2011 at Brookhaven National Laboratory of U.S.A. It is antihelium-4, a conglomeration of two antiprotons and two antineutrons. This antihelium is the antimatter partner to what physicists call the alpha particle (identical to a standard helium nucleus) and its discovery comes on the 100th anniversary of the discovery of the alpha particle itself. It will help to understand the evolution of the universe. It will also help probe the possibility of the existence of stars and galaxies made of antimatter.

37. In the context of which of the following do some scientists suggest the use of cirrus cloud thinning technique and the injection of sulphate aerosol into stratosphere?

- (a) Creating the artificial rains in some regions
(b) Reducing the frequency and intensity of tropical cyclones
(c) Reducing the adverse effects of solar wind on the Earth
(d) Reducing the global warming

I.A.S. (Pre) 2019

Ans. (d)

For reducing the global warming geo-engineering solutions have often been suggested by the scientists. Injection of sulphate aerosols, very fine solid particles, into the stratosphere is one of the most widely discussed ideas. Sulphate aerosol particles are very good reflectors of sunlight, and it has been shown, through various climate models, that even if 1% of current incident solar radiation is reflected back in space, a very significant amount of temperature rise on Earth can be offset. Another way of instantly cooling the planet is the use of cirrus cloud thinning technique. These clouds are composed mainly of ice crystals and they have greenhouse property. They let the solar radiation pass through and reach the surface, but trap the higher wavelength infrared

radiation emitted from the Earth, thereby contributing to the heating. If these clouds are reduced by some engineering interventions, it would allow the IR radiation from the Earth, too, to pass through to space, thus allowing some of the heat to dissipate and hence cool the planet.

38. Which one of the following chemicals had been in use till now for making artificial clouds ?

- (a) Potassium Nitrate (b) Heavy Water
(c) Sulphur Iodide (d) Silver Iodide

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (d)

Cloud seeding is the process of spreading either dry ice or more commonly silver aerosols into the upper part of clouds to try to stimulate the precipitation process and form rain. Tiny particles of silver iodide are sprayed on a cloud from an aeroplane. Sometimes, solid carbon dioxide (called dry ice) is used. The particles attract water droplets from the cloud. When they form a drop that is large enough, it starts raining.

39. The substance used for artificial rain is –

- (a) Silver bromide (b) Ammonium nitrate
(c) Silver iodide (d) All of the above

U.P.P.C.S. (Mains) 2008

Ans. (c)

See the explanation of above question.

40. Which of the following chemical is used in the artificial rain?

- (a) Silver Iodide (b) Potassium Chlorate
(c) Calcium Carbonate (d) Zinc Iodide

U.P.P.C.S. (Mains) 2013

Ans. (a)

See the explanation of above question.

41. Amount of rain depends on :

- (a) Air pressure
(b) Humidity in atmosphere
(c) Water cycle
(d) Temperature

U.P.P.C.S. (Pre) 2017

Ans. (b)

The amount of rain depends on the humidity in atmosphere. The amount of rainfall of all types such as drizzling, rainfall, snowfall etc. increase the amount of moisture in the atmosphere.

42. Match List-I with List-II and select the correct answer from the code given below the lists :

List-I (Substance)	List-II (Use)
A. Blue Vitriol	1. Artificial rain
B. Eosin	2. Fungicide
C. Silver iodide	3. Red ink
D. Zinc phosphide	4. Rodenticide

Code :

A	B	C	D
(a) 2	3	4	1
(b) 2	1	3	4
(c) 2	3	1	4
(d) 2	4	1	3

U.P. P.C.S. (Pre) 2020

Ans. (c)

The correctly matched lists are as follows :

List-I (Substance)	List-II (Use)
Blue Vitriol	– Fungicide
Eosin	– Red ink
Silver iodide	– Artificial rain
Zinc phosphide	– Rodenticide

43. Which one of the following pairs is correctly matched?

- (a) Silver Iodide – Horn Silver
(b) Silver Chloride – Artificial rain
(c) Zinc Phosphide – Rat poison
(d) Zinc Sulphide – Philosopher's wool

U.P.P.C.S. (Pre) 2011

Ans. (c)

The correctly matched pairs are as follows :

Silver Iodide	– Artificial rain
Silver Chloride	– Horn Silver
Zinc Phosphide	– Rat poison
Zinc Oxide	– Philosopher's wool

44. Match List-I with List-II and select the correct answer from the code given below the lists :

List-I	List-II
A. Kevlar	1. Explosive
B. Taxol	2. Synthetic fiber
C. Zinc Phosphide	3. Anticancer drug
D. Nitrocellulose	4. Rodenticide

Code :

A	B	C	D
(a) 2	4	3	1
(b) 2	3	4	1
(c) 4	3	1	2
(d) 3	1	2	4

U.P.P.C.S. (Pre) 2019

Ans. (b)

The correct match of List-I with List-II is as follows :

Kevlar	–	Synthetic fiber
Taxol	–	Anticancer drug
Zinc Phosphide	–	Rodenticide
Nitrocellulose	–	Explosive

45. Which one of the following is NOT correctly matched?

(Substance)	(Use)
(a) Silver bromide	– Artificial Photosynthesis
(b) Silver iodide	– Artificial rain
(c) Lithium bicarbonate	– Treatment of rheumatism
(d) Milk of magnesia	– Antacid

U.P.P.C.S. (Pre) 2019

Ans. (a)

Silver bromide is used in photography and it is not used in artificial photosynthesis. Pairs of other options are correctly matched.

46. Match List-I with List-II select the correct answer using the codes given below :

List-I	List - II
A. Silicon Carbide	Photosynthesis
B. Carbon Fibre	Refrigerator
C. Carbon dioxide	Artificial Diamond
D. Dichlorodifluoromethane	Aircraft

Code :

	A	B	C	D
(a)	4	3	1	2
(b)	3	4	1	2
(c)	2	3	1	4
(d)	3	2	1	4

U.P. Lower Sub. (Pre) 2009

Ans. (b)

Silicon Carbide	–	Artificial Diamond
Carbon Fibre	–	Aircraft
Carbon dioxide	–	Photosynthesis
Dichlorodifluoromethane	–	Refrigerator

47. Match List-I with List-II and select the correct answer by using the codes given below the lists :

List-I	List-II
A. Iron	1. Photography
B. Lead	2. Lightning conductor
C. Silver	3. Haemoglobin
D. Copper	4. Storage battery

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	2	1	4	3
(d)	3	4	1	2

U.P.P.S.C. (GIC) 2010

Ans. (d)

The main attribute common to all lighting rods is that they are all made of conductive materials such as copper and aluminium. Copper and its alloys are the most common materials used in lightning protection in buildings and houses. Haemoglobin in the blood carries oxygen from the respiratory organs (lungs or gills) to the rest of the body. Its molecule comprises four subunits, each containing an iron atom bound to a haem group. Lead-acid batteries, also known as lead storage batteries provide high current in short period of time, making them essential for vehicles. Silver Bromide is used in photography.

48. Match List-I with List-II and select the correct answer by using the codes given below the lists:

List-I	List-II
A. Morphine	1. Antiseptic
B. Sodium	2. Alloy
C. Boric Acid	3. Analgesic
D. German Silver	4. Kerosene oil

Code :

	A	B	C	D
(a)	4	3	1	2
(b)	2	4	3	1
(c)	3	1	4	2
(d)	3	4	1	2

U.P.P.C.S. (Pre) 2011

Ans. (d)

Morphine (INN) is an opioid analgesic drug. It is the main psychoactive chemical present in opium. Sodium is kept in kerosene oil because it is extremely reactive. It is kept under the oil because it might react with air. Boric acid is a weak monobasic Lewis acid of boron often used as an antiseptic, insecticide, flame retardant etc. German silver is a copper alloy with nickel and often zinc. The usual formulation is 60% copper, 20% nickel and 20% zinc. Nickel silver is named for its silvery appearance, but it contains no elemental silver unless plated.

49. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I	List-II
A. Skin Cancer	1. Chlorofluorocarbons
B. Noise Pollution	2. Ultraviolet light
C. Global Warming	3. Decibel
D. Ozone Hole	4. Carbon dioxide

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	4	1
(c)	2	3	1	4

- (d) 4 3 2 1

(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

The correct matching of List-I and List-II is as follows –

Skin Cancer	–	Ultraviolet Light
Noise Pollution	–	Decibel
Global Warming	–	Carbon dioxide
Ozone Hole	–	Chlorofluorocarbon

50. Which one of the following is called the 'metal of future'?

- (a) Copper (b) Iron
(c) Titanium (d) Aluminium

U.P.P.S.C. (GIC) 2010

Ans. (c)

Titanium is being increasingly used in our modern society. It is light, strong and corrosion-resistant. These properties allow it to be used in the aerospace industry, building industry, sports goods industry and as implants in a number of surgical procedures. Titanium is also called the 'metal of future'.

51. Which one of the following is NOT correctly matched?

- (a) Magnesium – Fireworks
(b) Uranium – Atom bomb
(c) Silver – Electric bulb
(d) Zinc – Galvanization

U.P.P.C.S. (Mains) 2002

Ans. (c)

Silver possesses the highest electrical conductivity, thermal conductivity and reflectivity of any metal.

The correctly matched pairs are as follows :

Magnesium	–	Fireworks
Uranium	–	Atom bomb
Tungsten	–	Electric bulb
Zinc	–	Galvanization

52. Chemically 'Vermillion' is –

- (a) Calcium Carbonate (b) Potassium Nitrate
(c) Mercuric Sulphide (d) Sodium Chloride

Uttarakhand P.C.S. (Pre) 2005

Ans. (c)

Vermillion is a brilliant red or scarlet pigment originally made from the powdered mineral cinnabar. It is chemically known as mercuric sulphide with chemical formula HgS. It is virtually insoluble in water.

53. Chemically vermillion is :

- (a) Calcium Carbonate (b) Potassium Nitrate

- (c) Potassium Sulphide (d) Mercury (II) Sulphide
(e) None of the above

Chhattisgarh P.C.S. (Pre) 2015

Ans. (d)

See the explanation of above question.

54. Which is used in Dry Washing?

- (a) Petrol (b) Benzene
(c) Alcohol (d) All the above
(e) None of these

Chhattisgarh P.C.S. (Pre) 2014

Ans. (d)

In spite of the name, dry-cleaning is not completely dry. Fluids are used in the dry-cleaning process. In the early days, garment scourers and dryers found several fluids that could be used as dry-cleaning solvents including camphene, benzene, alcohol, kerosene and gasoline. Thus, petrol, benzene and alcohol all are used in dry washing.

55. 'Red ink' is prepared from –

- (a) Phenol (b) Aniline
(c) Congo red (d) Eosin

U.P.R.O./A.R.O. (Pre) 2014

Ans. (d)

Eosin is a Tetrabromofluorescein. Most red inks are dilute solution of the red dye eosin.

56. Which one of the following colours is supplied by Sodium Chromate?

- (a) Blue (b) Red
(c) Green (d) Black

U.P. PCS (Mains) 2016

Ans. (b)

The Red colour is supplied by Sodium Chromate. Its chemical formula is Na₂CrO₄.

57. Which among the following would cause the bright red colour due to bursting of crackers?

- (a) Strontium (b) Sodium
(c) Sulphur (d) Magnesium
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Different elements and metal compounds/salts in the fire crackers impart different colours in bursting of them. Salts of barium give green colour whereas magnesium give a bright white colour. Salts of calcium produce orange, copper produce blue and sodium produce yellow/gold colour. The sparkling crimson red colour seen in firecrackers after they burst is due to the presence of strontium salts.

58. Match List-I with List-II and select the correct answer from the code given below the lists :

List-I	List-II
A. Acetone	1. Fire extinguisher
B. Carbon Tetrachloride	2. Nail polish remover
C. Hydrogen peroxide	3. Dressing of wounds
D. Liquid ammonia	4. Refrigerant

Code :

	A	B	C	D
(a)	4	3	2	1
(b)	3	4	1	2
(c)	2	1	3	4
(d)	1	3	4	2

U.P.R.O./A.R.O. (Mains) 2013

Ans. (c)

The correctly matched pairs are as follows :

Acetone	–	Nail polish remover
Carbon tetrachloride	–	Fire extinguisher
Hydrogen peroxide	–	Dressing of wounds
Liquid ammonia	–	Refrigerant

59. Which of the following pairs are not correctly matched–

- | | | |
|------------------------|---|------------------|
| (a) Potassium chlorate | – | Matchbox |
| (b) Manganese dioxide | – | Dry cell |
| (c) Hydrochloric acid | – | Accumulator cell |
| (d) Uranium | – | Atom bomb |

U.P. Lower Sub. (Pre) 1998

Ans. (c)

Sulphuric acid (not hydrochloric acid) is used in accumulator cell (Lead-acid cell). Other pairs are correctly matched.

60. Which of the following main elements is used in making matchbox?

- | | |
|--------------|-----------------|
| (a) Graphite | (b) Phosphorous |
| (c) Silicon | (d) Sodium |

U.P. Lower Sub. (Pre) 2004

Ans. (b)

The safety matchbox ignites due to the extreme reactivity of phosphorus with the potassium chlorate on the match head. Phosphorus is a highly reactive, non-toxic chemical used for making the striking surface of a matchbox.

61. Which one of the following is used for manufacturing safety matches?

- | | |
|--------------------|------------------------|
| (a) Di-phosphorus | (b) Black phosphorus |
| (c) Red phosphorus | (d) Scarlet phosphorus |

U.P. R.O./A.R.O. (Pre) 2016

Ans. (c)

Red phosphorus and phosphorus trisulfide is used for manufacturing safety matches. The matchstick's head is typically composed of the mixture of potassium chlorate, red lead, antimony sulphide and glue. The striking surface of matchboxes is typically composed of red phosphorus, antimony sulphide, powdered lead and glue.

62. When kept in air and darkness, which of the following elements glows spontaneously?

- | | |
|------------------------|-----------------------|
| (a) Red phosphorus | (b) White phosphorus |
| (c) Scarlet phosphorus | (d) Violet phosphorus |

U.P. P.C.S. (Mains) 2016

Ans. (b)

White phosphorus is a waxy, transparent solid. It sublimes if exposed to light. The sublimation is the process by which solid changes directly into a gas when heated, without first changing to a liquid. The White phosphorus is phosphorescent. It gives off a beautiful greenish–white glow.

63. Carbon Nanotubes (CNTs) were prepared by –

- | | |
|-------------|------------|
| (a) Fuller | (b) Iijima |
| (c) Faraday | (d) Raman |

U.P.P.C.S. (Mains) 2008

Ans. (b)

Sumio Iijima is a Japanese physicist, often cited as the inventor of carbon nanotubes.

64. In India, water desalination plant is located in –

- | | |
|-----------------|-------------|
| (a) Lakshadweep | (b) Goa |
| (c) Cuttack | (d) Chennai |

U.P.P.C.S. (Mains) 2007

Ans. (a) & (d)

In India, first water desalination plant was opened in 2005 at Kavaratti in the Lakshadweep islands. This LTTD (Low Temperature Thermal Desalination) plant with a capacity of 1 lakh litre per day was developed indigenously by National Institute of Ocean Technology (NIOT). Now, Chennai also have water desalination plants.

65. Where was the first desalination plant in India to produced one lakhs litres fresh water per day based on low temperature thermal desalination principle commissioned ?

- | | |
|---------------|----------------|
| (a) Kavaratti | (b) Port Blair |
| (c) Mangalore | (d) Valsad |

I.A.S. (Pre) 2008

Ans. (a)

See the explanation of above question.

66. Which chemical acts as a reclamation agent for alkaline soil?

- | | |
|-----------------------|----------------------|
| (a) Calcium Carbonate | (b) Calcium Sulphate |
|-----------------------|----------------------|

- (c) Calcium Oxide (d) None of the above

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

The term 'alkali soil' is used to designate land that is saline and has an excess of absorbed sodium carbonate and sodium chloride. An excess of sodium chloride gives a hard firm surface crust with cubical salt crystals. Alkali salts in the soil also affect the plant growth by reducing the size of leaves, plant roots remain superficial, bark of stem turns brown or black. Gypsum is calcium sulphate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) that has no direct effect on soil pH. The main use of Gypsum is the reclamation of highly alkaline soil. Gypsum dissolves to release calcium which replaces sodium and makes the soil fertile.

67. Which one of the following concerns soil?

- (a) Edaphic (b) Climatic
(c) Biotic (d) Topography

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (a)

Edaphic is related to soil. Edaphic's qualities may characterise the soil itself, including drainage, texture or chemical properties such as pH.

68. Liquid Oxygen or liquid Hydrogen is used technology in obtaining the industrial level called-

- (a) Pyrotechnics (b) Pyrometry
(c) Hydroponics (d) Cryogenics

U.P.P.S.C. (GIC) 2010

Ans. (d)

Cryogenics is the study and use of materials including biological products at very low temperature. Nitrogen, Hydrogen and Oxygen in liquid form are used as in many cryogenic applications. Another use of cryogenics is cryogenic fuels for rockets with liquid hydrogen is used as fuel in the rocket propellant while liquid oxygen (LOX) is used as an oxidizer.

69. Which one of the following was used as a chemical weapon in the First World War ?

- (a) Carbon monoxide (b) Hydrogen cyanide
(c) Mustard gas (d) Water gas

I.A.S. (Pre) 1997

Ans. (c)

Mustard gas or sulphur mustard ($\text{C}_4\text{H}_8\text{Cl}_2\text{S}$) is a chemical compound which has been used as a chemical weapon in the First World War. Its chemical name is 'Dichlorodiethyl Sulphide'. It is a poisonous gas. It can cause ulcers on the skin.

70. Which of the following is used in World War-I ?

- (a) Luciste (b) Mustard gas
(c) Methyl cyanide (d) Phosgene

U.P. Lower Sub. (Pre) 2009

Ans. (b)

Sulfur mustard, commonly known as mustard gas, is a chemical warfare agent which was first used effectively in World War-I by the German army against British and Canadian soldiers near Ypres, Belgium in 1915.

71. Mustard gas is used –

- (a) As a fuel gas
(b) In chemical warfare
(c) As a source of mustard oil
(d) As a deodorant

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

See the explanation of above question.

72. Poisonous mustard gas is a :

- (a) Gas (b) Liquid
(c) Solid (d) None of the above

Jharkhand P.C.S. (Pre) 2016

Ans. (b)

In room temperature the poisonous mustard gas is found in liquid state. It is used as a powerful chemical weapon. Skin, eyes and lungs are mostly affected by this toxic chemical. The symptoms of this chemical started to be visible after 1–6 hours.

73. Smoke screen is used in warfare for the purpose of concealment and camouflage. Smoke screens generally consists of fine particles of –

- (a) Sodium chloride dispersed in air
(b) Silver iodide dispersed in air
(c) Titanium oxide dispersed in air
(d) Magnesium oxide dispersed in air

U.P.P.C.S. (Mains) 2009

Ans. (c)

Smokescreens generally consist of fine particles of titanium oxides. Titanium dioxide is a naturally occurring oxide of titanium. A smoke screen is a cloud of smoke created to conceal military operations. Zinc chloride is also used in creating smoke screen.

74. “The final Agent Orange raid in Vietnam took place in 1970 - areas have begun to bloom again. But 19 years after the end of the war, it seems plain that Agent Orange is killing and maiming human beings, something it never intended to do. The apparent toxic fallout from those clouds is a crop of human miseries including cancers, miscarriages and birth defects that may persist for decades”. The offensive substance referred to in this quotation is :

- (a) DDT used as insecticide
(b) A complex mixture of herbicides and weedicides used to increase agricultural output in the South Vietnam under the US aid programme

- (c) A complex mixture of DDT and other insecticides used as aerial sprays for protection against malaria and other tropical diseases
- (d) Dioxin used as defoliants

I.A.S. (Pre) 1995

Ans. (d)

Agent Orange (or herbicide orange) is one of the herbicides and defoliants used by the U.S. military as part of its herbicidal warfare programme, Operation Ranch Hand during the Vietnam War from 1962 to 1971. Agent Orange, which contained the chemical dioxin, was the most commonly used of the herbicide mixtures and the most effective. It was later revealed to cause serious health issues including tumours, birth defects, rashes and cancer.

75. What is Agent Orange?

- (a) An ace spying group trained by the Police in London
- (b) A weed-killing chemical used by the US Military in the Vietnam War
- (c) A technique used in the advanced railway safety procedure
- (d) A special chemical used in making alloy steel

I.A.S. (Pre) 2008

Ans. (b)

See the explanation of above question.

76. Which one of the following is incorrect?

- (a) Theory of evolution was propounded by Charles Darwin.
- (b) The breaking apart of the nucleus of an atom is called fusion.
- (c) Dry ice is nothing but solid Carbon dioxide.
- (d) Telephone was invented by Graham Bell.

U.P.P.S.C. (GIC) 2010

U.P.Lower (Spl) (Pre) 2004

Ans. (b)

Nuclear fusion is an atomic reaction in which multiple atoms combine to create a single, more massive atom. The resulting atom has a slightly smaller mass than the sum of the masses of the original atoms. Nuclear fission is a reaction in which a large nuclei break apart into smaller nuclei, releasing a great amount of energy. In this way statement (b) is wrong, while other statements are correct.

77. In dry cell, which of the following are used as electrolytes?

- (a) Ammonium Chloride and Zinc Chloride
- (b) Sodium Chloride and Calcium Chloride
- (c) Magnesium Chloride and Zinc Chloride
- (d) Ammonium Chloride and Calcium Chloride

I.A.S. (Pre) 2009

Ans. (a)

A dry cell is a type of electrochemical cell, commonly used in the form of a battery in many electrical appliances, torch, calculator etc. Unlike a wet cell, a dry cell can operate in many orientations without spilling as it contains no free liquid. A common dry cell is a zinc-carbon battery, sometimes it is called the dry Leclanche cell. Ammonium chloride is used as the electrolyte. Sometimes ammonium chloride is replaced by zinc chloride for having high energy voltage.

78. In dry cell, which of the following are used as electrolytes?

- (a) Ammonium Chloride and Zinc Chloride
- (b) Ammonium Chloride and Calcium Chloride
- (c) Magnesium Chloride and Zinc Chloride
- (d) Sodium Chloride and Zinc Chloride

Jharkhand P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

79. In a common dry cell, electrolyte is –

- (a) Zinc
- (b) Sulphuric acid
- (c) Ammonium Chloride
- (d) Manganese dioxide

39th B.P.S.C. (Pre) 1994

Ans. (c)

See the explanation of above question.

80. Assertion (A) : A dry cell cannot be charged.

Reason (R) : Dry cell has a reversible reaction.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

U.P. Lower Sub. (Pre) 2004

Ans. (c)

The matters of a dry cell are there in dry form. A zinc pot is there having full of manganese dioxide, sal ammoniac and carbon. This cell cannot be charged because the chemical reaction in it cannot be reversed.

81. The energy, found in dry cell is –

- (a) Mechanic
- (b) Electric
- (c) Chemical
- (d) Electromagnetic

M.P.P.C.S. (Pre) 1992

Ans. (c)

The best example of changing into electric energy from chemical energy is primary cells or batteries, the dry cell is also an example of this phenomenon.

82. In a car battery, electrolyte substance used is-

- (a) Hydrochloric acid (b) Sulphuric acid
(c) Nitric acid (d) Distilled water

42nd B.P.S.C. (Pre) 1997

Ans. (b)

Electrolyte substance used in a car battery is sulphuric acid (H_2SO_4). It is a strong acid.

83. Which one of the following acids is used in the car-batteries?

- (a) Acetic Acid (b) Hydrochloric Acid
(c) Sulphuric Acid (d) Nitric Acid

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (c)

See the explanation of above question.

84. Which one of the following solutions of electrolytes is used in the car battery?

- (a) Sodium sulphate (b) Nitric acid
(c) Sulphuric acid (d) Potassium nitrate

U.P.P.C.S. (Pre) 2015

Ans. (c)

See the explanation of above question.

85. The acid used in a car battery is :

- (a) Acetic acid (b) Hydrochloric acid
(c) Nitric acid (d) Sulfuric acid
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

See the explanation of above question.

86. Which one of the following pairs of materials serve as electrodes in chargeable batteries commonly used in devices such as torchlights, electric shavers etc.?

- (a) Nickel and Cadmium (b) Zinc and Carbon
(c) Lead peroxide and Lead (d) Iron and Cadmium

I.A.S. (Pre) 1995

Ans. (a)

Nickel and Cadmium are mainly used as electrodes in rechargeable batteries of mobiles, torchlights and electric shavers etc. In this, cathode of nickel hydroxide and anode of cadmium is used. Potassium hydroxide is mainly used as electrolyte in it.

87. In microwave oven, the microwave tube used is –

- (a) A combination of klystron and magnetron tubes
(b) Klystron tube
(c) Magnetron tube
(d) Travelling wave tube

R.A.S./R.T.S.(Pre) 2012

Ans. (c)

The Magnetron tube is a high-powered vacuum tube which is used to produce high microwave energy. In this, the magnetic operation is based on the motion of electrons under the combined influence of electric and magnetic fields. For the tube to operate, electrons must flow from the cathode to anode. Nowadays these are used in microwave ovens.

88. Match List – I with List – II and select the correct answer from the code given below :

List –I

List - II

- | | |
|--------------------------------|-------------------------------|
| A. Freon | 1. Anti-knocking agent |
| B. Tetraethyl lead | 2. Refrigerant |
| C. Benzene hexachloride | 3. Fire extinguisher |
| D. Carbon tetrachloride | 4. Insecticide |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 2 | 1 | 4 |
| (c) | 4 | 1 | 2 | 3 |
| (d) | 2 | 1 | 4 | 3 |

U.P.P.C.S. (Mains) 2013

Ans. (d)

The correct match of list-I with list-II is given below :

Freon	–	Refrigerant
Tetraethyl lead	–	Anti-knocking agent
Benzene hexachloride	–	Insecticide
Carbon tetrachloride	–	Fire extinguisher

89. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I

List-II

(Substances)

(Applications)

- | | |
|---------------------|--------------------------------|
| A. Aspartame | 1. Synthetic rubber |
| B. Freon | 2. Anti-histamine |
| C. Neoprene | 3. Artificial Sweetener |
| D. Benadryl | 4. Refrigerant |

Code :

- | | | | | |
|-----|--|---|---|---|
| | A | B | C | D |
| (a) | 1 | 3 | 4 | 2 |
| (b) | 2 | 4 | 1 | 3 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 3 | 1 | 2 | 4 |
| (e) | None of the above/More than one of the above | | | |

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

The correct matching is as follows :

Aspartame	–	Artificial Sweetener
Freon	–	Refrigerant
Neoprene	–	Synthetic Rubber
Benadryl	–	Anti-histamine

90. Consider the following pairs :

Commonly used/ Consumed materials	Unwanted or controversial chemicals likely to be found in them
--	---

- | | | |
|----------------------|---|---------------------------|
| 1. Lipstick | - | Lead |
| 2. Soft drinks | - | Brominated vegetable oils |
| 3. Chinese fast food | - | Monosodium glutamate |

Which of the pairs given above is/are correctly matched?

- | | |
|------------------|------------------|
| (a) 1 only | (b) 2 and 3 only |
| (c) 1 and 3 only | (d) 1, 2 and 3 |

I.A.S. (Pre) 2017

Ans. (d)

Lead is a common impurity found in cosmetic products, such as lipsticks, eye shadows, blushes, compact powders, shampoos and body lotions etc. Brominated vegetable oil is a food additive used primarily to help emulsify citrus-flavored soft drinks, preventing them from separating during distribution. Monosodium glutamate (MSG) is a glutamate or salt of glutamic acid, a 'non-essential' amino acid. It is a neurotransmitter—transporting messages from one nerve cell to another. Because it is said to enhance flavours, some scientists believe it 'excites nerve endings' and exhibits 'neuro-excitatory properties,' that the ability to stimulate neurons. It is used in Chinese fast food. Hence, all three pairs are correctly matched.

91. Match List-I with List-II and select the correct answer using the code given below the lists :

- | List-I | List-II |
|---------------------------|----------------|
| A. Potassium bromide | 1. Fertilizer |
| B. Potassium nitrate | 2. Photography |
| C. Potassium sulphate | 3. Bakery |
| D. Monopotassium tartrate | 4. Gunpowder |

Code :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 2 | 4 | 1 | 3 |
| (b) 2 | 4 | 3 | 1 |
| (c) 4 | 2 | 3 | 1 |
| (d) 4 | 2 | 1 | 3 |

I.A.S. (Pre) 1998

Ans. (a)

Potassium bromide is used in photography.
Potassium nitrate is used in gunpowder.
Potassium sulphate is used as fertilizer.
Monopotassium tartrate is used in the bakery.

92. Which one of the following pairs is correctly matched?

- | | | |
|------------------------|---|--------------------|
| (a) Sulphur dioxide | — | Teeth |
| (b) Fluoride pollution | — | Bhopal gas tragedy |

- | | | |
|-----------------------|---|-------------|
| (c) Methyl isocyanate | — | Acid rain |
| (d) Ozone depletion | — | Skin cancer |

U.P.C.S. (Pre) 2015

Ans. (d)

The correctly matched Pairs are :

- | | | |
|--------------------|---|--------------------|
| Sulphur dioxide | — | Acid rain |
| Fluoride pollution | — | Teeth |
| Methyl isocyanate | — | Bhopal gas tragedy |
| Ozone depletion | — | Skin cancer |

Thus it is clear that option (d) is correctly matched.

93. Assertion (A) : Phenyl is used as a household germicide.

Reason (R) : Phenyl is phenol derivative and phenol is an effective germicide.

- | |
|---|
| (a) Both (A) and (R) are true, and (R) is the correct explanation of (A). |
| (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A). |
| (c) (A) is true, but (R) is false. |
| (d) (A) is false, but (R) is true. |

I.A.S. (Pre) 1998

Ans. (a)

Phenyl is an effective germicide. It is a derivative of phenol, that is why it also has germicidal property. Actually phenyl is an emulsion of light creosote oil and water with soap. Creosote oil contains carbolic acid, creosol and other homologues of phenol which exert the necessary germicidal power.

94. Match List-I (Scientists) and List-II (Inventions) on the basis of code –

- | List-I | List-II |
|-----------------|--------------|
| A. Rutherford | 1. Powerloom |
| B. Alfred Nobel | 2. Telephone |
| C. Cartwright | 3. Dynamite |
| D. Graham Bell | 4. Atom Bomb |

Code :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 1 | 2 | 3 | 4 |
| (b) 4 | 3 | 1 | 2 |
| (c) 3 | 4 | 2 | 1 |
| (d) 4 | 2 | 1 | 3 |

Chhattisgarh P.C.S. (Pre) 2008

Ans. (b)

The correctly matched lists are as follows :

- | Scientists | Inventions | |
|--------------|------------|-----------|
| Rutherford | — | Atom Bomb |
| Alfred Nobel | — | Dynamite |
| Cartwright | — | Powerloom |
| Graham Bell | — | Telephone |

95. Match List – I with List – II and select the correct answer from the code given below :

List –I (Specification)	List - II (Date)
A. Technology Day	1. 9 August
B. Photography Day	2. 29 August
C. National Sports Day	3. 11 May
D. Nagasaki Day	4. 19 August

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	4	3	2	1
(c)	4	3	1	2
(d)	3	4	1	2

U.P.P.C.S. (Mains) 2003

Ans. (a)

The correctly matched lists are as follows :

Technology Day	–	11 May (Pokharan test 1998)
Photography Day	–	19 August
National Sports Day	–	29 August
		(Major Dhyanchand Jayanti)
Nagasaki day	–	9 August
		(Nagasaki - Japan atom bomb)

96. For which branch of science U.N. has designated the year 2011 ?

- The International Year of Chemistry
- The International Year of Astronomy
- The International Year of Biodiversity
- The International Year of Physics

Uttarakhand P.C.S. (Pre) 2010

Ans. (a)

The year 2011 was declared the International Year of Forests and International Year of Chemistry by the United Nations.

2023	–	International Year of Millets
	–	International Year of Dialogue as a Guarantee of Peace
2024	–	International Year of Camelids
2025	–	International Year of Peace and Trust
	–	International Year of Glaciers' Preservation
	–	International Year of Cooperatives
	–	International Year of Quantum Science and Technology
2026	–	International Year of Rangelands and Pastoralists
	–	International Year of Volunteers for Sustainable Development
	–	International Year of the Woman Farmer
2027	–	International Year of Sustainable and Resilient Tourism
2029	–	International Year of Asteroid Awareness and Planetary Defence.

97. The United Nations Framework Convention on Climate Change deals with

- reduction in fossil fuel usage
- CO₂ emissions mitigation
- reduction in Uranium production
- greenhouse gases emissions mitigation
- None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

The United Nations Framework Convention on Climate Change deals with greenhouse gases emission mitigation. It is known fact that CO₂ falls under greenhouse gases. On 22 April, 2016, 175 countries/organisations including India signed the historic 'Paris Climate Change Agreement' to reduce carbon emission. The participating countries had agreed that the average temperature of 21st century should not be increased for more than 2°C in comparison to the temperature of pre-industrial era.

98. Consider the following pairs :

International	Subject
agreement / set-up	
1. Alma-Ata Declaration	– Healthcare of the people
2. Hague Convention	– Biological and chemical weapons
3. Talanoa Dialogue	– Global climate change
4. Under2 Coalition	– Child rights

Which of the pairs given above is/are correctly matched?

- 1 and 2 only
- 4 only
- 1 and 3 only
- 2, 3 and 4 only

I.A.S. (Pre) 2020

Ans. (c)

Alma-Ata Declaration was the first international declaration underlining the importance of primary healthcare, which was adopted in 1978 at the International Conference on Primary Healthcare held in Almaty (formerly Alma-Ata), Kazakhstan. **Hague Convention** (1980) on the civil aspects of international child abduction is a multilateral treaty that provides an expeditious method to return a child internationally abducted by a parent from one member country to another. **Talanoa Dialogue** was a process designed to help countries implement and enhance their Intended Nationally Determined Contributions (INDCs : reductions in greenhouse gas emissions under the UNFCCC) by the end of 2020. **Under 2 Coalition** is a global community of state and regional governments committed to ambitious climate action in line with the Paris Agreement. It is a coalition of subnational governments, founded in 2015, that aims to achieve greenhouse gases emissions mitigation. Hence, only pairs 1 and 3 are correctly matched.

99. Which of the following countries has made it mandatory to blend ethanol with gasoline?

- (a) China (b) U.K.
(c) Switzerland (d) Brazil

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (d)

Since 1976, the Brazilian Government has made it mandatory to blend ethanol with gasoline and since 2007, the legal blend is around 25% ethanol and 75% gasoline.

100. Which of the following is an inorganic gaseous pollutant of air?

- (a) Phosgene (b) Hydrogen Cyanide
(c) Hydrogen Sulphide (d) Methane

R.A.S./R.T.S. (Pre) (Re. Exam) 2013

Ans. (b) & (c)

Among the given options, Hydrogen Cyanide and Hydrogen Sulphide both are inorganic gaseous pollutants of air. Some other inorganic pollutants are Carbon dioxide, Carbon monoxide, Sulphur dioxide, Nitrous oxide and Nitrogen dioxide.

101. The chief source of Hydrogen Sulphide pollutant is :

- (a) Decaying vegetation and animal matter
(b) Automobiles
(c) Oil Refineries
(d) Thermal Power Plants

U.P. R.O./A.R.O. (Pre) 2021

Ans. (a)

Hydrogen sulphide (H_2S) is a colourless gas with the odour of rotten eggs. H_2S is naturally formed during decaying vegetation and animal matter and bacterial decomposition of human and animal wastes, and is present in emissions from sewage treatment facilities and landfills. These are the chief environmental sources of H_2S pollutant. H_2S is also released naturally from geothermal fields. Its industrial sources include oil and natural gas extraction and petrochemical plants, coke oven plants, and kraft paper mills.

102. Which gas contributes to global warming through its greenhouse effect?

- (a) Carbon dioxide (b) Nitrous oxide
(c) Methane (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (d)

Carbon dioxide, nitrous oxide and methane – all these gases are greenhouse gases and contribute to global warming through greenhouse effect.

103. The relative contributions of carbon dioxide (CO_2), nitrous oxide (N_2O), chlorofluorocarbons (CFCs) and methane (CH_4) towards global warming are :

- (a) $CO_2 > CH_4 > CFCs > N_2O$
(b) $CO_2 > N_2O > CH_4 > CFCs$
(c) $CO_2 > CFCs > CH_4 > N_2O$
(d) $CO_2 > CH_4 > N_2O > CFCs$

R.A.S./R.T.S. (Pre) 2016

Ans. (d)

The relative contribution of main greenhouse gases towards global warming are as follows :

- (i) Water vapour – about 60%
(ii) Carbon dioxide (CO_2) – about 26%
(iii) Methane (CH_4) – about 4-9%
(iv) Ozone (O_3) – about 3-7%
(v) Nitrous oxide (N_2O) – about 1.5%
(vi) Chlorofluorocarbons (CFCs) and others – 0.1-0.4%

Thus, it is clear that option (d) is correct.

104. With reference to 'water vapour', which of the following statements is/are correct?

1. It is a gas, the amount of which decreases with altitude.
2. Its percentage is maximum at the poles.
Select the answer using the code given below:
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2024

Ans. (a)

Water vapour is a variable gas in the atmosphere, the amount of which decreases with altitude. In the warm and wet tropics, it may account for about four percent of the air by volume, while in the dry and cold areas of desert and polar regions, it may be less than one percent of the air. Water vapour also decreases from the equator towards the poles. It also absorbs parts of the insolation from the sun and preserves the Earth's radiated heat. It thus, acts like a blanket allowing the Earth neither to become too cold nor too hot. Water vapour also contributes to the stability and instability in the air. Hence, statement 1 is correct, while statement 2 is incorrect.

105. Harnessing of nuclear energy often causes :

- (a) Air Pollution (b) Water Pollution
(c) Thermal Pollution (d) Noise Pollution

U.P. R.O./A.R.O. (Mains) 2017

Ans. (c)

Thermal pollution is the degradation of water quality by any process that changes the water temperature. A common cause of thermal pollution is the use of water as coolant in power plants (thermal as well as nuclear power plants). When water used as a coolant is returned to the natural environment at a higher temperature, the sudden change in temperature decreases oxygen supply and affects ecosystem composition.

106. 'Green Muffler' is used against which type of pollution?

- (a) Air (b) Water
(c) Soil (d) Noise

U.P. P.C.S. (Mains) 2017

Ans. (d)

Green mufflers are barriers grown near noisy places to reduce the impact of noise. This is a technique of reducing noise pollution by planting 4-5 rows around the populated areas or noisy places like along roadsides, industrial areas, societies near highways etc. These dense trees reduce noise pollution as they filter out the noise and obstruct it reaching the citizens.

107. Magnetic particles, suspected to cause neurodegenerative problems, are generated as environmental pollutants from which of the following?

1. Brakes of motor vehicles
2. Engines of motor vehicles
3. Microwave stoves within homes
4. Power plants
5. Telephone lines

Select the correct answer using the code given below.

- (a) 1, 2, 3 and 5 only (b) 1, 2 and 4 only
(c) 3, 4 and 5 only (d) 1, 2, 3, 4 and 5

I.A.S. (Pre) 2021

Ans. (b)

Magnetite is an oxide formed by iron ions of valences 2+ and 3+. Biologically formed nanoparticles of the strongly magnetic mineral, magnetite, were first detected in the human brain over 25 years ago. These brain magnetites display compelling similarity with the magnetite nanospheres formed by combustion, which are ubiquitous and prolific in urban, airborne particulate matter (PM). Magnetite can have potentially large impacts on the brain due to its unique combination of redox activity, surface charge, and strongly magnetic behaviour. Magnetite could be toxic to the brain, generating harmful reactive oxygen species that have been linked to Alzheimer's disease. In addition to occupational settings (including, for example, exposure to printer toner powders), higher concentrations of magnetite pollution nanoparticles are likely to arise from open fires and vehicles (tailpipe and brake wear) and/or industrial (power plants) PM sources. There are no clear evidence of generation of magnetic particles from microwave stoves within homes or from telephone lines. Hence, option (b) is the correct answer.

108. The incomplete burning of petrol and diesel produces :

- (a) nitric oxide (b) nitrogen dioxide
(c) carbon dioxide (d) carbon monoxide
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

In the complete combustion (burning) of hydrocarbon fuels (i.e. petrol and diesel), hydrocarbons react with oxygen and water and carbon dioxide are produced. Incomplete combustion occurs when the supply of air or oxygen is poor. In the incomplete combustion of hydrocarbon fuels water is still produced, but carbon monoxide (a poisonous gas) and carbon (as soot) are produced instead of carbon dioxide.

109. The main air pollutant(s) in metropolitan cities is/are

- (a) O₃ (b) CO and SO₂
(c) CO₂ and NO₂ (d) None of the above

M.P. P.C.S. (Pre) 2018

Ans. (*)

In metropolitan cities, emitted smoke from automobiles and industries contains carbon monoxide, carbon dioxide, oxides of nitrogen, sulphur dioxide, lead, ozone etc. which are present in all options. Thus, the commission did not consider this question in evaluation.

110. What is particulate matter?

- (a) Solid residue (b) Air pollutant
(c) Water pollutant (d) Soil pollutant
(e) None of the above/more than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Particulate matter (PM) also known as particle pollution, is a complex mixture of extremely small particles and liquid droplets that get into the air. Once inhaled, these particles can affect the heart and lungs and cause serious health problems.

111. The size of the Respirable Suspended Particulate Matter (R.S.P.M.) in air is :

- (a) more than 7 µ m (micron)
(b) more than 6 µ m (micron)
(c) more than 5 µ m (micron)
(d) less than 5 µ m (micron)

U.P.P.C.S. (Pre) 2017

Ans. (d)

R.S.P.M. is that fraction of Total Suspended Particulate Matter (T.S.P.M.) which is readily inhaled by humans through their respiratory system and in general considered as particulate matter with their diameter less than 2.5 microns (PM 2.5).

112. Under Government of India's National Air Quality Monitoring Programme (NAMP) covering 312 cities/towns in 29 States and 6 Union Territories of the country, which of the following combination has been identified for regular monitoring at all the locations?

- (a) Sulphur Dioxide (SO₂), Nitrogen Dioxide (NO₂), Carbon Monoxide (CO), and Suspended Particulate Matter (PM₁₀)
(b) CO, NO₂, PM₁₀ and PM_{2.5}

- (c) CO, NO₂, Oxygen Difluoride (OF₂) and PM_{2.5}
 (d) Sulphur Dioxide (SO₂), Nitrogen Dioxide (NO₂), Suspended Particulate Matter (PM₁₀) and Fine Particulate Matter (PM_{2.5})

Jharkhand P.C.S. (Pre) 2021

Ans. (d)

Central Pollution Control Board (CPCB) is executing a nationwide programme of ambient air quality monitoring known as National Air Quality Monitoring Programme (NAMP). The network at present consists of 931 operating stations covering 398 cities/towns. The objectives of the N.A.M.P. are to determine status and trends of ambient air quality; to ascertain whether the prescribed ambient air quality standards are violated; to Identify Non-attainment Cities; to obtain the knowledge and understanding necessary for developing preventive and corrective measures and to understand the natural cleansing process undergoing in the environment through pollution dilution, dispersion, wind based movement, dry deposition, precipitation and chemical transformation of pollutants generated. Under N.A.M.P., earlier (in the question period) four air pollutants viz., Sulphur Dioxide (SO₂), Oxides of Nitrogen (as NO₂), Respirable Suspended Particulate Matter (RSPM/PM₁₀) and Fine Particulate Matter (PM_{2.5}) were identified for regular monitoring at all the locations. At Present, only first three among these have been identified for regular monitoring at all the locations. The monitoring of meteorological parameters such as wind speed and wind direction, relative humidity (RH) and temperature were also integrated with the monitoring of air quality.

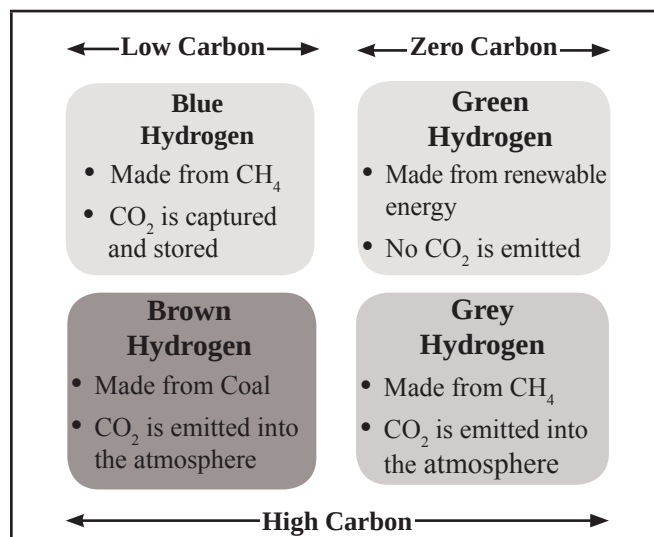
113. Steam methane reforming technique is used to obtain:

- (a) Brown Hydrogen (b) Grey Hydrogen
 (c) Green Hydrogen (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (b)

Steam methane reforming (SMR) is a widely used industrial process for producing hydrogen and carbon monoxide (syngas) by reacting methane (natural gas) with steam at high temperatures (700 – 1000°C) in the presence of a catalyst, typically nickel-based. The main purpose of this technology is often hydrogen production, although syngas has multiple other uses such as production of ammonia or methanol. The main chemical reaction is : $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$. To further increase hydrogen production, the carbon monoxide produced in the SMR reaction can be reacted with steam in a "water-gas shift reaction" : $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$. SMR is a primary method for hydrogen production, used in various industries, including fuel cells, ammonia synthesis, and the production of other chemicals. Hydrogen produced by steam reforming is termed as 'grey' hydrogen when the waste carbon dioxide is released to the atmosphere and 'blue' hydrogen when the carbon dioxide is (mostly) captured and stored geologically. Hence, option (b) is the correct answer.



114. Consider the following heavy industries :

1. Fertilizer plants
2. Oil refineries
3. Steel plants

Green Hydrogen is expected to play a significant role in decarbonizing how many of the above industries?

- (a) Only one (b) Only two
 (c) All three (d) None

I.A.S. (Pre) 2023

Ans. (c)

Green Hydrogen, produces using renewable energy, has the potential to play a key role in low-carbon and self-reliant economic pathways. Green Hydrogen can enable utilization of domestically abundant renewable energy resources across regions, seasons, and sectors, feeding multiple usage streams, either as a fuel or as an industrial feedstock. It can directly replace fossil fuel derived feedstocks in petroleum refining, fertilizer production, steel manufacturing etc. Hydrogen fuelled long-haul automobiles and marine vessels can enable decarbonisation of the mobility sector. In the recent years, pilot projects have been undertaken in India for production of Green Hydrogen through electrolysis of water using renewable electricity, and from biomass through thermochemical and biochemical routes. The 'National Green Hydrogen Mission' aims to provide a comprehensive action plan for establishing a Green Hydrogen ecosystem and catalysing a systemic response to the opportunities and challenges of this sunrise sector. The Mission aims to develop and scale up Green Hydrogen production technology and make it affordable and widely accessible. Hence, option (c) is the correct answer.

115. With reference to green hydrogen, Consider the following statements :

- 1. It can be used directly as a fuel for internal combustion.**
- 2. It can be blended with natural gas and used as fuel for heat or power generation.**
- 3. It can be used in the hydrogen fuel cell to run vehicles.**

How many of the above statements are correct?

- (a) Only one (b) Only two
(c) All three (d) None

I.A.S. (Pre) 2023

Ans. (c)

Green Hydrogen can be consumed through either direct combustion, electricity generation through fuel cells, or industrial processes to be used as chemical feedstock. Direct use includes industrial processes in iron and steel plants and refineries; transportation fuel for light duty vehicles, buses, trucks, trains, and potentially shipping and aircrafts; and power sector storage and grid balancing and for co-firing in thermal power plants. It can also be used in the hydrogen fuel cell to run vehicles. Petroleum and Natural Gas Regulatory Board (PNGRB), the regulatory body has given approval for 5% vol./vol. blending of green hydrogen with PNG to start with and the blending level would be scaled phase wise to reach 20%. Green hydrogen when blended with natural gas reduces CO₂ emissions keeping net heating content same. Recently NTPC Ltd. commissioned India's first green hydrogen blending project. Hence, all three statements are correct.

116. Smog is essentially caused by the atmospheric presence of

- (a) Oxygen and ozone
(b) Ozone and nitrogen
(c) Oxygen and nitrogen
(d) Oxide of nitrogen and sulphur

U.P.P.C.S. (Pre) 2019

Ans. (d)

Smog is a type of intense air pollution. The word 'smog' is a contraction of the words smoke and fog to refer the smoky fog. It is a kind of visible air pollution and is composed of nitrogen oxides, sulphur oxides, ozone, smoke and other particulates. Man-made smog is derived from coal combustion emissions, vehicular emissions, industrial emissions, forest and agricultural fires and photochemical reactions of these emissions.

117. Photochemical smog always contains :

- (a) Ozone (b) Aluminium ion
(c) Phosphorus (d) Methane

U.P. P.C.S. (Pre) 2022

Ans. (a)

Photochemical smog is a brownish-gray haze caused by the action of solar ultraviolet radiation on atmosphere polluted with hydrocarbons and oxides of nitrogen. It contains anthropogenic air pollutants, mainly ozone, nitric acid, and organic compounds, which are trapped near the ground by temperature inversion. Ozone is the principal component of air pollution of this kind. Photochemical smog appears to be initiated by nitrogen oxides that are emitted into the air as pollutants mainly from internal combustion engines. Absorbing the visible or ultraviolet energy of sunlight, it forms nitric oxide (NO) to free atoms of oxygen (O), which then combine with molecular oxygen (O₂) to form ozone (O₃). In the presence of hydrocarbons (other than methane), certain other organic compounds, and sunlight, various chemical reactions take place to form photochemical smog – a mixture of ozone, nitric acid, aldehydes, peroxyacyl nitrates (PANs) and other secondary pollutants.

118. With reference to Eutrophication, which of the following statements is/are correct?

- 1. It is a phenomenon of nutrient enrichment of a water body.**
- 2. It depletes dissolved oxygen of the water.**

Select the correct answer using the code given below :

Code :

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

U.P.P.C.S. (Pre) 2020

U.P.P.C.S. (Pre) 2019

Ans. (c)

Eutrophication is excessive richness of nutrients in a lake or other body of water, frequently due to run-off from the land, which causes a dense growth of plant life. This induces excessive growth of algae and this process depletes dissolved oxygen of the water. Eutrophication is often induced by the discharge of nitrate or phosphate-containing detergents, fertilizers, or sewage into an aquatic system.

119. Eutrophication of water body results from –

- (a) Submersing idols in the water body.
(b) Depletion of oxygen.
(c) Enhanced growth of algae or algal blooms.
(d) High amounts of nitrogenous nutrients and orthophosphates.

R.A.S./R.T.S. (Re. Exam) (Pre) 2013

Ans. (d)

Eutrophication of bodies of water is a naturally occurring phenomenon. However, the process has been aggravated by the human population. Such man-made eutrophication is caused by the excessive discharge of nutrients, especially phosphorous (P) in the form of phosphate, nitrogen (N) in the form of nitrate, and silicate.

120. Biological Oxygen Demand (BOD) is a standard criterion for –

- (a) Measuring oxygen levels in blood
- (b) Computing oxygen levels in forest ecosystems
- (c) Pollution assay in aquatic ecosystems
- (d) Assessing oxygen levels in high altitude regions

I.A.S. (Pre) 2017

Ans. (c)

Biological Oxygen Demand (BOD) is a standard criterion for pollution assay in aquatic ecosystems. BOD is the amount of oxygen consumed by bacteria and other microorganisms while they decompose organic matter under aerobic conditions. Bacteria and other microorganisms consume dissolved oxygen in water bodies to decompose organic matter or waste, which reduces the amount of dissolved oxygen in water. The decrease in the amount of dissolved oxygen means increase in its demand. Higher BOD indicates more oxygen is required, signifying lower or infected water quality. Therefore, the demand for oxygen is directly related to the amount of increasing waste. Where there is high BOD, there will be low dissolved oxygen.

121. Physico-chemical characteristics of water in water sources undergo changes due to –

- (a) Aquatic macrophytes
- (b) Aquatic fungi
- (c) Effluents
- (d) Evapotranspiration

I.A.S. (Pre) 1996

Ans. (c)

Physico-chemical characteristics of water in water sources undergo changes due to effluents. Effluent is generally considered to be water pollution, such as the outflow from a sewage treatment or wastewater from discharge of industries. The outflow of polluted water is mainly done in rivers, lake, sea, etc.

122. 'Membrane Bioreactors' are often discussed in the context of:

- (a) Assisted reproductive technologies
- (b) Drug delivery nanotechnologies
- (c) Vaccine production technologies
- (d) Wastewater treatment technologies

I.A.S. (Pre) 2024

Ans. (d)

A Membrane Bioreactor (MBR) is a chemical equipment that combines a membrane separation process with a conventional wastewater treatment process. MBRs are commonly used in municipal and industrial wastewater treatment plants (WWTPs). In the MBR process, membranes act as a solid-liquid separation device, keeping the biomass within the bioreactor before discharging the treated effluent to the nature. Basically, they take the place of bulky clarifiers used in the conventional activated sludge (CAS) process.

123. In the context of which one of the following are the terms 'pyrolysis and plasma gasification' mentioned?

- (a) Extraction of rare earth elements
- (b) Natural gas extraction technologies
- (c) Hydrogen fuel-based automobiles
- (d) Waste-to-energy technologies

I.A.S. (Pre) 2019

Ans. (d)

Like incineration, pyrolysis and plasma gasification technologies are thermal processes that use high temperatures to break down waste. The main difference is that they use nil or less oxygen than traditional mass-burn incineration. Plasma gasification system facilitates safe disposal of all type of organic wastes. The inorganic waste materials like glass, metals, plastics can also be minimized to slag. In plasma pyrolysis the most likely compounds which are produced include carbonaceous matter, methane, carbon monoxide, hydrogen, carbon dioxide and water molecules. The product gases recovered has calorific and commercial values. These gases can be used as a heat source or as fuel. NITI Ayog in his three-year Action Agenda (2017-2020), also assessed the benefit-cost ratio of thermal pyrolysis and plasma gasification as waste-to-energy technologies.

124. Mercury is a toxic substance that severely contaminates land, water, air and the food chain. Which among the following is/are source(s) of mercury pollution?

- I. Pesticides**
- II. Dental amalgam fillings**
- III. Fluorescent lamp**
- IV. Coal-based thermal power plants**

- (a) Only I
- (b) Only I and II
- (c) Only I, II and III
- (d) I, II, III and IV all

Chhattisgarh P.C.S. (Pre) 2020

Ans. (d)

Pesticides, Dental amalgam fillings, fluorescent lamp and coal-based thermal power plants, all of these are sources of mercury pollution.